Student Fashion Preferences and Spending Analysis

23CSE452 – Business Analytics

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Abstract

This project analyzes student fashion purchasing behavior from a **business analytics perspective**. It examines how students make fashion-related decisions, including their preferred clothing styles, shopping methods, spending habits, and the factors that influence their trust in fashion brands. The study is based on a dataset of **5,006 student survey responses**, providing valuable insights into the purchasing patterns of the student demographic, which represents a significant segment of the fashion retail market.

Data preprocessing and **exploratory data analysis (EDA)** were conducted to identify key spending trends and shopping preferences among students. In addition, **Natural Language Processing (NLP)** techniques such as TF-IDF were applied to analyze open-ended feedback, enabling the identification of important factors influencing customer satisfaction and purchase regret. **Machine learning models** were also developed to predict students' preferred shopping methods, helping businesses better understand consumer behavior.

The findings reveal that students strongly prefer an **omnichannel shopping approach**, combining both online and offline retail experiences. The analysis also highlights that **price sensitivity, product quality, and clothing fit** are major factors influencing student purchasing decisions. These insights can help fashion retailers design more effective **pricing strategies, targeted marketing campaigns, and improved customer experiences** to better serve the student market segment.

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1 Introduction

The fashion industry is constantly changing, especially when it comes to college students. Students are an important group of customers for fashion brands because they actively follow new trends and styles. However, their fashion choices are often influenced by many factors such as social media, friends, seasonal trends, and most importantly, their budget. Because of these changing preferences, it can be difficult for fashion brands to clearly understand what students actually like and how much they are willing to spend.

When brands do not fully understand student preferences, several problems can occur. For example, companies may introduce products that do not match student expectations, which can lead to confusion while making purchase decisions. This may also result in dissatisfaction after buying products and higher return rates. In addition, marketing strategies may become less effective if they are based on assumptions instead of real customer data. With the rapid growth of online shopping platforms, it has also become important for businesses to understand whether students prefer shopping online, offline, or using both methods.

This project aims to address these challenges by using **business analytics techniques** to study the fashion buying behavior of college students. By analyzing students' clothing preferences, shopping methods, and spending habits, the study attempts to generate useful insights that can help fashion retailers better understand their target customers and make more informed business decisions.

2 Problem Statement

In the highly competitive fashion retail market, understanding customer behavior is essential for making effective business decisions. College students represent an important consumer segment for fashion brands, yet their purchasing behavior is influenced by multiple factors such as trends, social media, peer influence, and budget limitations. Due to these constantly changing preferences, many retailers struggle to accurately understand student fashion choices and their preferred shopping channels.

Without proper data-driven insights, businesses often rely on assumptions when developing products, setting prices, or planning marketing strategies. This can lead to mismatched product offerings, ineffective promotional campaigns, lower customer satisfaction, and increased product return rates. With the rapid growth of digital commerce, it has also become important for retailers to understand whether students prefer **online shopping, offline shopping, or a combination of both (omnichannel behavior)**.

The key problem addressed in this project is how **business analytics techniques** can be used to analyze student fashion preferences, spending patterns, and shopping behavior in order to generate meaningful insights. By applying data preprocessing, ex-

ploratory data analysis, text mining, and machine learning models, the project aims to **identify patterns in student purchasing behavior and predict preferred shopping methods.**

The insights derived from this analysis can help fashion retailers make **data-driven business decisions**, improve customer segmentation, design targeted marketing strategies, optimize pricing models, and enhance the overall shopping experience for the student market.

3 Dataset Selection and Understanding

The dataset used in this project was collected through a Google Forms survey titled **"Student Fashion Preferences and Spending Analysis."** The survey was designed to gather information about students' clothing preferences, shopping habits, and spending patterns. Since the data was collected directly from students, it provides a realistic view of how college students make fashion-related purchasing decisions.

The dataset contains **5,006 responses and 18 variables**. Each row in the dataset represents the response of one student, while each column represents a specific survey question related to fashion preferences or shopping behavior.

During preprocessing, the **Timestamp column was removed** because it was not relevant for analyzing student fashion preferences and shopping behavior.

Some of the important variables in the dataset include:

- Gender
- Level of Study
- Preferred Clothing Type
- Shopping Method Preference
- Average Annual Spending
- Price Range Preference
- Purchase Frequency
- Open-ended Feedback

From a **Business Analytics perspective**, this dataset is very useful because it contains both **structured data** (such as categories and numerical values) and **unstructured data** (text feedback). This allows the use of different analytics techniques to better understand student behavior.

For example, **descriptive analytics** helps in understanding patterns such as how much students spend on fashion and how often they shop. **Diagnostic analytics** helps identify the reasons behind certain behaviors, such as why students regret purchases or what factors influence their choices. **Predictive analytics** can be used to build machine learning models that predict students' preferred shopping methods.

Overall, this dataset provides a strong base for applying business analytics techniques to study student fashion behavior and generate insights that can help fashion retailers make better business decisions.

4 Tools and Technologies

The analysis for this project was conducted using the Python programming language in a Jupyter Notebook environment. Several data science libraries were used to perform data preprocessing, visualization, text analysis, and machine learning tasks.

The main tools and libraries used include:

Pandas – for data manipulation and preprocessing

NumPy – for numerical computations

Matplotlib and Seaborn – for data visualization

NLTK – for natural language processing and sentiment analysis

Scikit-learn – for machine learning models, PCA, and clustering

MLxtend – for Apriori algorithm and association rule mining

5 Methodology

The analysis for this project was carried out using **Python in a Jupyter Notebook environment.** Several data science libraries such as **Pandas, Matplotlib, Seaborn, NLTK, and Scikit-Learn** were used for data processing, visualization, text analysis, and machine learning.

The methodology followed a step-by-step business analytics workflow.

5.1 Data Preprocessing

The dataset was cleaned by removing unnecessary columns such as the timestamp field and handling missing values. Text data was standardized, and spending ranges were converted into numerical values to make the data suitable for analysis.

5.2 Exploratory Data Analysis

EDA was performed to understand patterns in student fashion behavior.

```
Dataset Loaded Successfully!
  Total Samples: 5006
  Total Columns: 18
```

	Timestamp	Gender	What is your current level of study?	Which type of clothing do you prefer the most?	What do you usually prefer to wear for college use?	Does your college have a uniform policy?	How often you buy clothes ?	Which shopping method do you prefer?	what do you prefer in this for shopping ?	What price range do you prefer per clothing item?	What is your average spending ?\n(per year)\n	Are discounts and offers important while purchasing clothes?	How confident are you while selecting clothes without assistance?	What is the main reason for regret ?	Do you tak opinion befor purchasing if ye whos opinion
0	2025-12-19 07:40:29	Female	MBA	Casual wear	Mixed	Yes	3-6 Months	Offline shopping	Better prices, Try before buying	₹500 – ₹1000	8000.0	Yes	Very confident	Price not worth it	No on
1	2025-12-19 08:10:51	Female	B.Tech / B.E	Casual wear	Casual wear	No	2-3 Weeks	Both	Better prices, More variety, Try before buying	Above ₹2000	15000.0	Sometimes	Very confident	Different from images	Reviews rating
2	2025-12-19 08:58:58	Male	B.Tech / B.E	Casual wear	Mixed	Yes	Once a Year	Online shopping	More variety, Convenience	₹500 – ₹1000	5000.0	Yes	Very confident	No regret	Famil
3	2025-12-19 09:22:03	Male	Diploma	Formal wear	Jeans & T-shirts	No	Once a Year	Both	More variety, Convenience	₹500 – ₹1000	20000.0	Yes	Somewhat confident	Quality issues	Famil
4	2025-12-19 09:27:03	Female	MBBS/BDS	Casual wear	Formals	Yes	1-2 Months	Both	Better prices, Try before buying	₹500 – ₹1000	8000.0	Yes	Very confident	Poor fit	Friend

Figure 1: Sample Data set

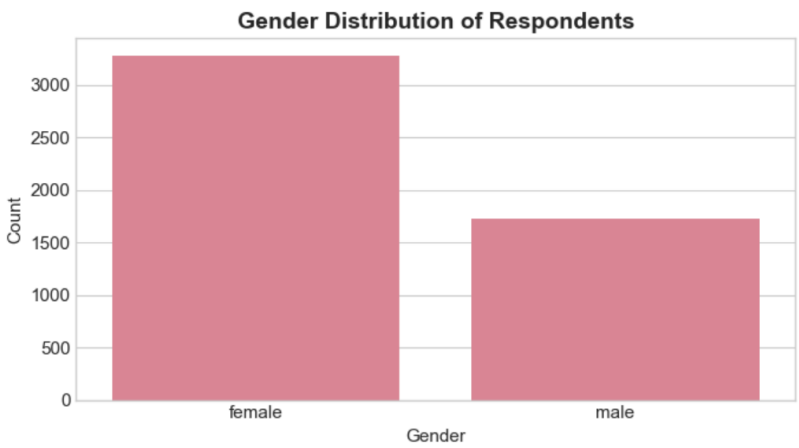

Figure 2: Gender Distribution

Visualizations were created to analyze spending habits, clothing preferences, and preferred shopping methods.

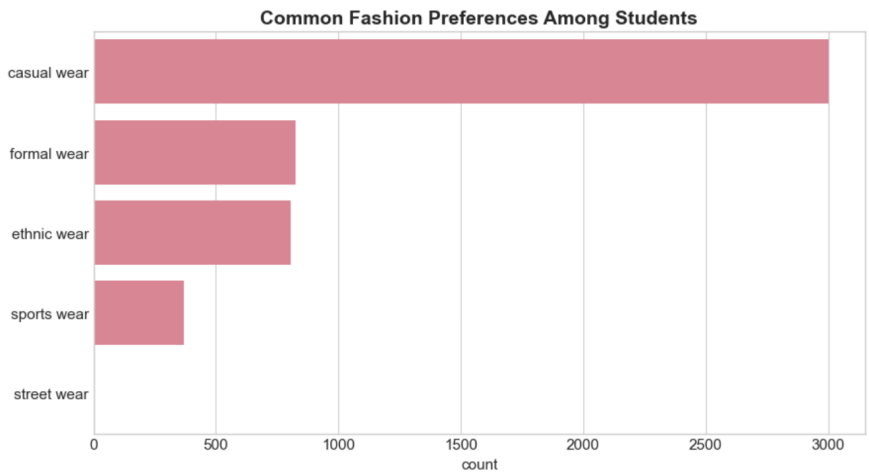

Figure 3: Clothing Preference

Figure 4: Shopping Method Preference

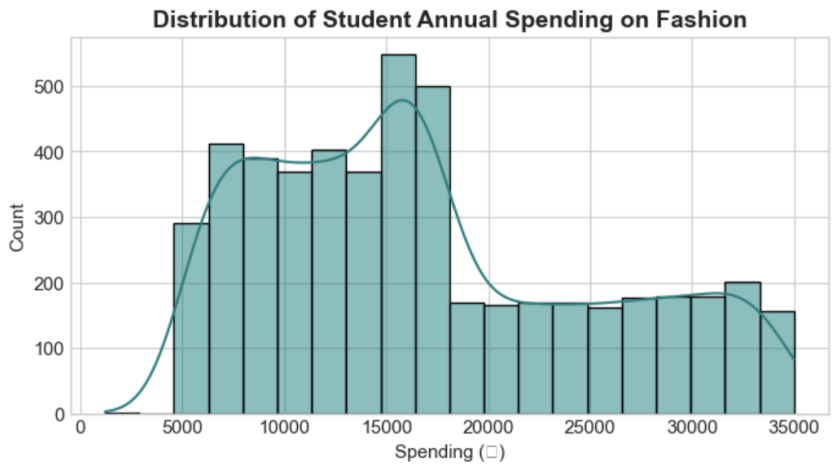

Figure 5: Spending Distribution

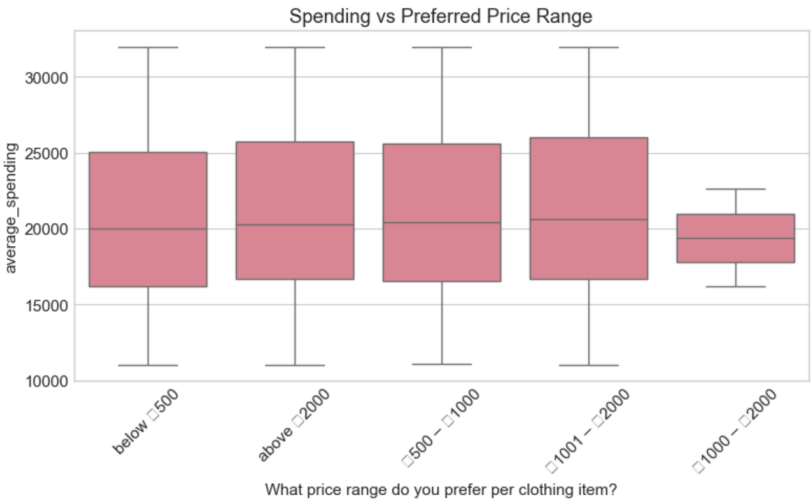

Figure 6: Spending vs Preferred Price Range

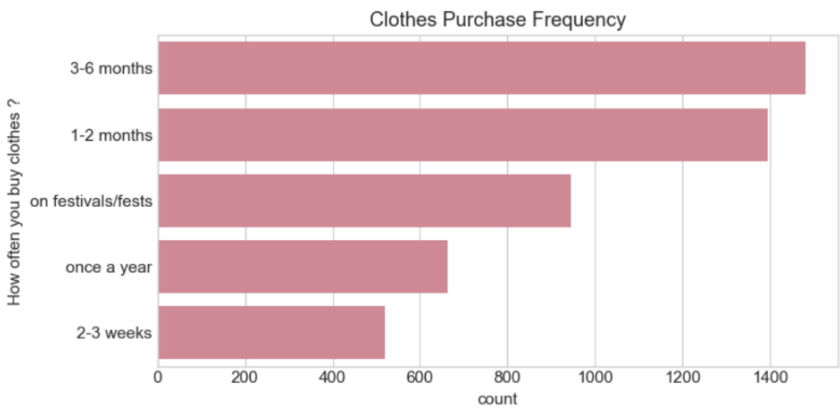

Figure 7: Purchase Frequency

This step helped identify key trends in how students shop and what factors influence their choices.

5.3 Text Mining

The open-ended feedback provided by students was analyzed using Natural Language Processing (NLP) techniques. The text data was cleaned through **tokenization, stop-word removal, and lemmatization**. After preprocessing, **TF-IDF vectorization** was applied to identify the most important keywords appearing in student feedback.

A **word cloud visualization** was also generated to highlight frequently occurring words related to fashion purchasing experiences and student satisfaction.

Figure 8: Key Words

5.4 Sentiment Analysis

Sentiment analysis was performed on student feedback using the VADER sentiment analyzer from the NLTK library. The feedback was classified into positive, negative, and neutral sentiments based on polarity scores. This analysis helped understand overall student satisfaction and identify common concerns related to fashion purchases.

5.5 Machine Learning Modeling

The dataset was divided into training and testing sets to evaluate model performance using accuracy metrics

The dataset was split into training and testing sets to evaluate model performance. Classification models including Naive Bayes, Logistic Regression, Support Vector Machine, Random Forest, and Gradient Boosting were trained to predict students' preferred shopping methods. Model performance was evaluated using accuracy scores and classification metrics. Among the tested models, **Support Vector Machine** achieved the highest accuracy, indicating its effectiveness for this dataset.

```
FINAL MODEL PERFORMANCE RANKING:
                Model  Accuracy
          SVM (Linear)  0.865135
           Naive Bayes  0.864136
   Logistic Regression  0.860140
     Gradient Boosting  0.860140
         Random Forest  0.850150
```

Figure 9: Accuracy Ranking

From a **business perspective**, this methodology helps convert raw survey data into meaningful insights. By combining data analysis, text mining, and predictive modeling, the study enables retailers to better understand student preferences, identify key factors influencing purchase decisions, and design more effective marketing and retail strategies.

5.6 Dimensionality Reduction using PCA

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset and visualize patterns in student fashion preferences. PCA transforms the original variables into a smaller set of components that capture the most important information in the data. This helps simplify the dataset while retaining the key relationships between variables.

By projecting the dataset into a two-dimensional space using the first two principal components, it becomes easier to observe patterns in student shopping behavior. The visualization also helps analyze how different shopping preferences such as online, offline, and both channels are distributed across the dataset.PCA helps visualize the relationships between multiple fashion-related features by projecting them into a lower-dimensional space.

The PCA visualization helps identify clusters and relationships between variables such as spending behavior, shopping preference, and clothing choices.

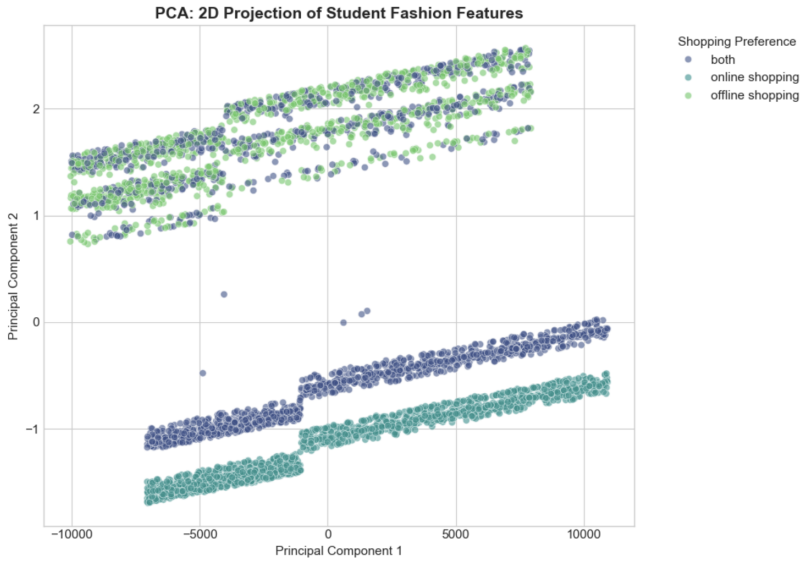

Figure 10: PCA 2D Projection of Student Fashion Features

5.7 Student Segmentation using K-Means Clustering

To further analyze student purchasing behavior, K-Means clustering was applied to segment students into different groups based on their fashion preferences and spending patterns. Clustering is an unsupervised machine learning technique that groups similar data points together without predefined labels.

The K-Means algorithm identified distinct clusters of students with different spending behaviors. For example, the segmentation analysis revealed groups of students with lower spending levels, moderate spending levels, and higher spending levels. These clusters represent different customer segments within the student market.

Visualizing the clusters in the PCA space makes it easier to interpret the segmentation results. Each cluster represents a group of students with similar fashion purchasing patterns.

From a business perspective, this segmentation can help fashion retailers design targeted marketing strategies, personalized promotions, and product offerings for different student groups.

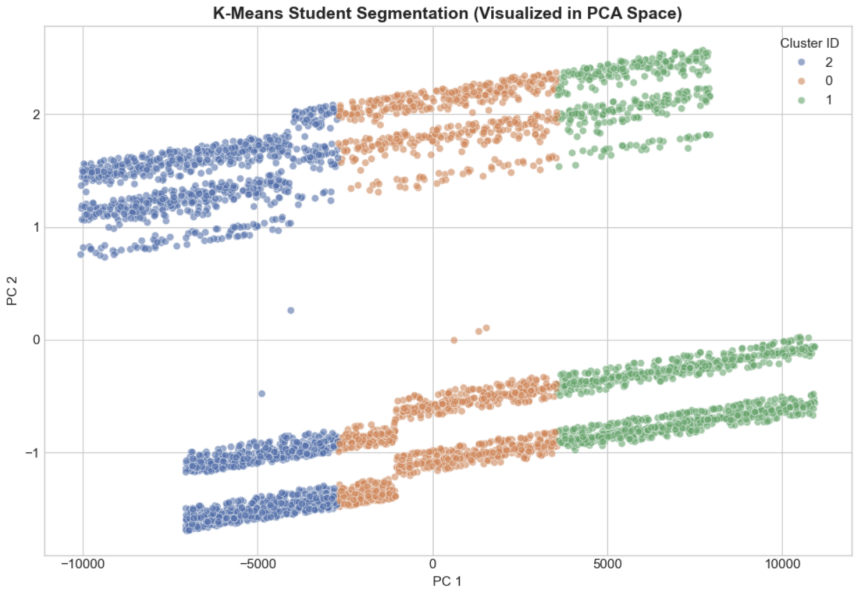

Figure 11: Student Segmentation using K-Means Clustering (Visualized in PCA Space)

5.8 Market Basket Analysis using Apriori

To identify relationships between different fashion preferences, association rule mining was applied using the Apriori algorithm. This technique helps discover frequently occurring combinations of items within the dataset. Association rules were generated to understand patterns in student purchasing behavior and fashion choices. These insights can help retailers design bundled offers and targeted product recommendations.

6 Business Insights

The analysis of the student fashion dataset using **exploratory data analysis, text mining, and machine learning techniques** produced several important insights for fashion retailers. These insights help businesses better understand student purchasing behavior and provide guidance for designing more effective marketing, pricing, and retail strategies.

6.1 Omnichannel Shopping Preference

The analysis showed that a large number of students prefer **using both online and offline shopping channels** instead of relying on only one method. This indicates that students value the convenience of online shopping while still appreciating the experience of visiting physical stores.

From a business perspective, this highlights the importance of adopting **omnichannel retail strategies.** Retailers should integrate their digital platforms with physical stores

by offering services such as online ordering with in-store pickup, easy return policies, and consistent pricing across channels. This approach can improve customer satisfaction and strengthen brand loyalty among student customers.

As shown in **Figure 4**, a majority of students prefer using both online and offline shopping channels.

6.2 Price Sensitivity

The exploratory data analysis revealed that most students spend **between 5,000 and 15,000 per year on fashion products**. This indicates that students are highly price sensitive and usually make purchasing decisions based on affordability.

For businesses, this insight suggests that **competitive pricing, student discounts, seasonal offers, and promotional campaigns** play an important role in attracting student customers. Retailers targeting the student market should focus on value-based pricing strategies and offer budget-friendly product options.

Figure 5 indicates that most students spend between 5,000 and 15,000 annually on fashion purchases.

6.3 Product Fit and Quality Concerns

The analysis of open-ended feedback using **text mining and TF-IDF techniques** highlighted common issues faced by students during fashion purchases. Many students mentioned problems such as **poor clothing fit, low product quality, and differences between product images and the actual items received.**

These issues often lead to dissatisfaction and increased product return rates. To address this, retailers should improve **size guides, provide detailed product descriptions, and use accurate product images or videos**. Improving these aspects can reduce customer dissatisfaction and build stronger trust with buyers.

6.4 Importance of Reviews and Ratings

Another key insight from the survey responses is that students rely heavily on **online reviews and ratings** before making purchasing decisions. Instead of traditional advertising, students tend to trust feedback from other customers who have already purchased the product.

This means businesses should actively encourage **customer reviews, ratings, and user-generated content** on their platforms. Displaying authentic customer feedback helps build trust and influences purchasing decisions among student consumers.

6.5 Predictive Insights from Machine Learning

The machine learning models used in the project were trained to **predict students preferred shopping methods** based on various features such as demographics, spending behavior, and clothing preferences. Among the tested models, the **Support Vector Machine Classifier showed the best performance**, indicating its effectiveness in identifying patterns in student shopping behavior.

From a business analytics perspective, such predictive models can help retailers **anticipate customer preferences and design targeted marketing strategies.** By understanding which students are more likely to prefer online or offline shopping, businesses can personalize recommendations and optimize their marketing efforts.

6.6 Customer Segmentation Insight

Clustering analysis identified different groups of students based on their spending behavior and shopping preferences. These clusters represent different types of customers within the student market. Understanding these segments can help fashion retailers design targeted promotions and product offerings for different customer groups.

6.7 Business Recommendations

Based on the analysis, several recommendations can be made for fashion retailers targeting student customers. Retailers should adopt omnichannel strategies that integrate online platforms with physical stores to improve convenience. Offering student discounts, seasonal promotions, and affordable pricing options can help attract price-sensitive students. Retailers should also focus on improving product quality, accurate size guides, and detailed product descriptions to reduce dissatisfaction and product returns. Additionally, encouraging customer reviews and ratings can build trust and positively influence purchasing decisions.

7 Conclusion

This project demonstrates how **business analytics techniques can be effectively applied to understand consumer behavior in the fashion retail industry**, particularly within the student demographic. By using data preprocessing, exploratory data analysis, text mining, and machine learning models, the study was able to analyze student fashion preferences, spending habits, and shopping behaviors using real survey data.

The analysis revealed several important insights. Students generally prefer an **omnichannel shopping experience**, combining both online and offline shopping. The

study also showed that students are **highly price sensitive**, with most purchases influenced by discounts and affordable price ranges. In addition, issues related to **clothing fit, product quality, and mismatch between product images and actual items** were identified as common reasons for dissatisfaction. The results also highlighted that **online reviews and ratings play a significant role in building trust and influencing purchase decisions** among students.

From a business perspective, these findings suggest several important solutions for fashion retailers. Businesses should adopt **omnichannel retail strategies** that seamlessly integrate online platforms with physical stores to improve customer convenience. Retailers should also focus on **competitive pricing, student discounts, and promotional campaigns** to attract budget-conscious consumers. Improving **product transparency, accurate size guides, and high-quality product descriptions** can help reduce return rates and increase customer satisfaction. Additionally, encouraging **authentic customer reviews and user-generated content** can strengthen brand trust among student buyers.

Overall, this project shows that **data-driven decision making through business analytics can help fashion retailers better understand their target market, improve marketing strategies, and enhance customer experience**. By leveraging these insights, businesses can develop more effective strategies to capture and retain the growing student fashion market.